

Group 9 Proposal: Block / Layer-Level Redundancy

Project “Redunformer” – Redundancy in Large Language Models, Summer Semester 2026

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We study redundancy at the granularity of a full transformer block (attention + FFN + residual). Our Weeks 1–2 baseline pipeline (GPT-2 $\times$ WikiText-2 perplexity, with HellaSwag via lm-eval-harness) is already implemented; we build measurement and intervention on top of it. The plan is training-free and one-shot: measure block redundancy, remove the least important blocks, compare against a random-removal baseline, with short fine-tuning as a recovery extension.

1. **Definition of redundancy.** A block is redundant if its output hidden state is nearly identical to its input, so skipping it leaves downstream computation almost unchanged. We quantify this with the *Block Influence* (BI) score of Men et al. (2024): one minus the average input–output cosine similarity of a layer,

$$BI_i = 1 - \mathbb{E}_{X,t} \left[\frac{X_{i,t}^\top X_{i+1,t}}{\|X_{i,t}\|_2 \|X_{i+1,t}\|_2} \right],$$

where $X_{i,t}$ is the hidden state of token t entering block i . Low BI $\Rightarrow$ redundant.

2. **Most relevant work.** Primary method: *ShortGPT* (Men et al., 2024) – the BI metric plus direct layer removal, training-free. Contrast: *LayerDrop* (Fan et al., 2019) – structured dropout during training that makes a network robust to inference-time layer removal; this motivates our recovery (fine-tuning) experiments. We additionally survey follow-ups (LaCo, SLEB) to position our method.
3. **Model(s) and hardware.** Primary: **GPT-2 small** (`gpt2`, 124M, 12 layers), matching our working baseline. The loader is device-agnostic (CUDA > MPS > CPU); the baseline was first verified on Apple Silicon (MPS), and redundancy experiments run on a consumer GPU (8–16 GB). **GPT-2 medium** (355M, 24 layers) is our model-size extension – it needs only an un-hardcoded model name, since loading is already generic. 12 layers gives coarse depth resolution, but a careful small-model study fits the seminar’s guidance.
4. **Datasets and metrics.** Primary metric: **WikiText-2 (raw) perplexity**, sliding-window (context 1024, stride 512), already implemented; current baseline ≈ 25.2 . Downstream: **HellaSwag** via lm-eval-harness (wired up), with LAMBADA and ARC-Easy as planned, more sensitive additions. At this scale multiple-choice accuracy is a weak signal, so perplexity carries most of the information. Every run logs model, dataset, metric, seed (42), device, and command to JSON, satisfying the reproducibility requirement.
5. **Dependence of redundancy.** *Depth*: test whether ShortGPT’s pattern (middle–late layers redundant, first and last critical) holds across our 12 layers. *Component*: block-level hidden states cannot separate attention from FFN; we treat this as a limitation, with an optional extension that computes BI over the `.attn` and `.mlp` sub-layers via forward hooks. *Model size*: GPT-2 small vs. medium. *Task*: perplexity vs. HellaSwag accuracy.
6. **Measurement plan.** GPT-2’s forward pass already exposes every layer’s hidden state by setting `output_hidden_states=True`: the $(L+1)$ returned states are each block’s input and output, so BI needs no custom hooks and reuses the existing `eval.py` forward loop over the WikiText-2 calibration windows. We compute (a) per-layer BI (bar chart); (b) an $L \times L$ CKA / cosine heatmap over all layer pairs, to reveal structure beyond adjacent layers; (c) the residual-update norm $\|X_{i+1} - X_i\|$ as a cross-check. New code lives in a `metrics/` module and `scripts/run_measurement.py`, matching the shared repo layout.
7. **First intervention. One-shot layer removal** (no retraining): delete the k lowest-BI blocks by slicing `model.transformer.h` and updating `config.n_layer`, sweep k , and plot perplexity / accuracy vs. removal ratio against a **random-removal baseline** at the same k . Later weeks add short fine-tuning recovery and neighbour-layer replacement (`pruning/`, `recovery.py`).
8. **Expected limitations.** GPT-2 small is the smallest, least-redundant regime; ShortGPT removes $\approx 25\%$ of layers at $\approx 92\%$ performance on 7B+ models, so we expect steeper degradation here – testing whether the redundancy claim survives at 124M is itself part of our question. Depth resolution is coarse (12 layers); block granularity cannot isolate attention vs. FFN without hooks; `float16` inference may add minor numerical noise (we may switch to `float32` for measurement); and GPT-2 differs architecturally from LLaMA (learned positional embeddings, no RoPE/GQA), limiting cross-family generalization.

References. Men et al. (2024), *ShortGPT: Layers in LLMs are More Redundant Than You Expect*, arXiv:2403.03853. Fan et al. (2019), *Reducing Transformer Depth on Demand with Structured Dropout*, arXiv:1909.11556.